

# Kyle Wong

Recent Master's graduate from Sidney Sussex College, University of Cambridge

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GitHub: <https://github.com/kyleyhw>

Website: [https://kyleyhw.github.io/personal\\_website/](https://kyleyhw.github.io/personal_website/)

## EDUCATION

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**Sidney Sussex College, University of Cambridge**, Cambridge, England, United Kingdom 2024 – 2025  
MASt/Part III in Astrophysics  
Graduated with Good Honours equivalent

**Victoria College, University of Toronto**, Toronto, Ontario, Canada 2020 – 2024  
HBSc in Physics & Mathematics  
Graduated with High Distinction

**German Swiss International School**, Hong Kong SAR, China 2010 – 2020  
International Baccalaureate (IB) Diploma in 6 subjects

## RESEARCH EXPERIENCE

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**Master of Advanced Study Research Project** Cambridge, England, United Kingdom  
*Postgraduate Student* October 2024 – September 2025

- Implemented and statistically validated variable cosmological initial conditions in the 21cmSPACE simulation package, enabling efficient forecasting of hydrogen distributions for upcoming radio astronomy experiments including the Square Kilometre Array (SKA).

**Canadian Institute for Theoretical Astrophysics**  
**Summer Undergraduate Research Fellowship** Toronto, Ontario, Canada  
*Undergraduate Researcher* May 2023 – December 2023

- Estimated neutron star tidal deformability from LIGO gravitational wave data using high-dimensional MCMC parameter estimation, and contributed new equation-of-state models to the collaboration's analysis pipeline.

**McGill Space Institute Summer Undergraduate Research Award** Montreal, Quebec, Canada  
*Undergraduate Researcher* May 2022 – April 2023

- Explored integration of statistical priors into the power spectrum estimator for the Hydrogen Epoch of Reionization Array (HERA) collaboration's cosmic dawn experiment data pipeline, improving signal inference from radio astronomy observations.

## FEATURED PERSONAL PROJECTS

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**mercury\_strategies** [Private Repository] (work in progress)

- Developed an event-driven engine for capturing alpha in weather-derivative prediction markets (Polymarket), managing a dockerised infrastructure on AWS with integrated risk management. Features a provider-agnostic collection and analysis layer, asynchronous watcher architecture for real-time signal generation, and automated execution with velocity rotation, local orderbook tracking, and risk-aware position sizing via the Kelly Criterion. Includes 4 months of manual prediction market trading to gain market knowledge, intuition, and experience.

**quant\_core** ([https://github.com/kyleyhw/quant\\_core](https://github.com/kyleyhw/quant_core))

- Developed an open-core algorithmic trading platform with a flexible, modular architecture for strategy development and automated execution via Interactive Brokers (IBKR). Features a comprehensive backtesting engine with high-efficiency data caching and integrated risk management.

**agent—evolve** (<https://github.com/kyleyhw/agent-evolve>)

- Created a multi-model, cooperative evolutionary code-optimisation agent skill. Supervisor, explorer, and reviewer roles can each be filled by any LLM through CLI, with candidates evolving on isolated git branches. Features interactive visualisation and surfaces the winner as a pull request for human approval.

## HONOURS, AWARDS AND SCHOLARSHIPS

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|-----------------------------------------------------------------------------------------------------------|------------------|
| • Dean's List Scholar in the Faculty of Arts & Science                                                    | 2022, 2023, 2024 |
| • Canadian Institute for Theoretical Astrophysics<br>Summer Undergraduate Research Fellowship (CITA SURF) | 9,500 CAD, 2023  |
| • Birkenshaw Family Scholarship II                                                                        | 1,000 CAD, 2023  |
| • McGill Space Institute (now Trottier Space Institute)<br>Summer Undergraduate Research Award (MSI SURA) | 7,000 CAD, 2022  |
| • David and Louise Fraser Scholarship                                                                     | 2,500 CAD, 2022  |
| • University of Toronto Scholar                                                                           | 1,500 CAD, 2022  |
| • Birkenshaw Family Scholarship                                                                           | 1,000 CAD, 2022  |

## SELECTED COURSES

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|-------------------------------------------------------------------|---------------------------------------------------|
| • Gravitational Waves & Numerical Relativity<br>(Cambridge, 2025) | • Relativity Theory I (Toronto, 2023)             |
| • Astrostatistics (Cambridge, 2025)                               | • Physics Project (Toronto, 2023)                 |
| • Cosmology (Cambridge, 2024)                                     | • Computational Physics (Toronto, 2023)           |
| • General Relativity (Cambridge, 2024)                            | • Computational Astrophysics (Toronto, 2023)      |
| • General Relativity (Toronto, 2024)                              | • Advanced Classical Mechanics (Toronto, 2023)    |
| • Relativity Theory II (Toronto, 2024)                            | • Geometry of Curves and Surfaces (Toronto, 2023) |
|                                                                   | • Partial Differential Equations (Toronto, 2023)  |

## SKILLS

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- **Technical:** Scientific computing, machine learning, statistical inference, simulation, time-series analysis, data visualisation, and high-performance computing, primarily in Python within Linux environments.
- **Financial Experience:** Trading in prediction markets and Counter-Strike skins.
- **AI Tools:** Claude Code, Google Gemini, Gemini CLI, ChatGPT, and Google Antigravity for automated software engineering, academic and scientific research, and coding.

## LANGUAGES

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- English (fluent, language of education)
- Cantonese (fluent, mother tongue)
- Mandarin (formally learned 10 years)
- German (formally learned 10 years)

## RECENT MEMBERSHIPS

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- Sidney Sussex College Football Team, Premier Division
- Institute of Astronomy Football Club
- Cambridge University Mountaineering Club
- Cambridge University Table Tennis Club

## PERSONAL INTERESTS

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- Rock climbing (9+ years experience both indoors and outdoors, with awards won at amateur, inter-high school and inter-university competitions. Also led a rock climbing extra-curricular activity during high school, including captaining the competition team)
- Soccer (7+ years league participation)
- Scuba diving (PADI open water certified)
- Travelling